Network Slicing and Stand Alone 5G Architectures: Paving the Way for Smart Cities and IoT

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***Abstract*— 5G networks represent a paradigm shift from their 4G predecessors, engineered to cater to the burgeoning demands of densely populated areas and the high data throughput requirements of the Internet of Things (IoT) and smart city infrastructures. Beyond enhancing fundamental network attributes such as reliability, availability, and security, 5G introduces architectural innovations to accommodate these advanced needs. This paper delves into the architectural nuances of 5G, focusing primarily on the Standalone (SA) deployment method, distinguished by its utilization of network slicing. This approach leverages cutting-edge virtual network technologies, including Software-Defined Networking (SDN) and Network Function Virtualization (NFV), to forge virtual network slices tailored to diverse consumer needs. Such customization is imperative for meeting the expansive and varied connectivity demands characteristic of 5G networks.To navigate the complexities of 5G architecture, this study advocates for the adoption of Reference Architectures (RAs), which provide a high-level abstraction framework for dissecting and understanding intricate systems. By proposing a model that outlines the standalone architecture of 5G networks, our research lays the groundwork for evolving this model into a comprehensive RA for 5G. This foundational work is instrumental in equipping system designers with the tools necessary for generalizing systems to manage design and deployment challenges effectively, as well as the heterogeneity inherent in system components. Additionally, our proposed model facilitates the establishment of a common vocabulary, promoting a coherent understanding of component interactions and offering a unified system perspective. Ultimately, this paper contributes a pivotal model that not only enhances our comprehension of 5G network architecture but also paves the way for future advancements in network design and deployment, promising to significantly impact the development of smart cities and IoT ecosystems.Keywords: 5G, 5G architecture, Reference architecture, patterns. Network slicing, Network Function Virtualization, Virtual Network Function.**

Keywords: 5G ,Non-Standalone (NSA) 5G Networks,Reference Architecture (RA),Network Slicing, Software-Defined Networking (SDN), Network Function Virtualization (NFV), Radio Access Network (RAN),Internet of Things (IoT)

I. INTRODUCTION

The evolution of telecommunications has been marked by significant technological advancements and expansive deployment. Notably, the advent of 4G technology has empowered users with low-latency capabilities, facilitating real-time interactions in video conferencing and online gaming. Theoretically, 4G networks can achieve bandwidths of up to 150Mbps [1]. Nonetheless, the introduction of 5G technology has ushered in a new era of data transmission speeds, capable of delivering data at rates hundreds of times faster than its predecessor, with potential speeds reaching up to 10Gbps. However, the benefits of 5G extend beyond mere speed enhancements. This technology promises minimal latency and increased network capacity, making it ideal for critical applications where latency could have severe repercussions, such as in Vehicular ad hoc networks (VANETs) and smart city infrastructures [2].

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To support the concurrent hosting of thousands of devices, an innovative technology known as network slicing has been developed. Network slicing allows for the creation of multiple logical network instances on a single physical network infrastructure. Network Function Virtualization (NFV) emerges as a pivotal technology in enabling network slicing through the virtualization of network functions, akin to how virtual servers operate with multiple virtual machines on a single physical hardware set. This concept is further extended in networks through the application of various network virtualization technologies, including Software Defined Networks (SDN) and NFV. A key advantage of network slicing is its capacity to tailor a unique, standalone network slice for specific use cases, thereby addressing the diverse service requirements of 5G applications, such as those in energy, infotainment, and healthcare. These sectors would not be efficiently served by the traditional, one-size-fits-all network architectural approach[3].

This paper aims to provide an exhaustive overview of the 5G architecture, enhancing understanding through the design of a Reference Architecture (RA) for the network with deep focus on the technology of network slicing. An RA is defined as an abstract software architecture that applies across one or more domains, without specific implementation details[4]. The structure of this paper is as follows: Section 2 presents a concise discussion on the concept of network slicing within 5G networks. Section 3 offers a background overview of two pivotal technologies contributing to the 5G architecture, namely SDN and NFV. Section 4 proposes a suggested Reference Architecture for 5G networks. Lastly, Section 5 concludes the paper with a synthesis of the findings.

II. 5G NETWORK SLICING

To achieve its ambitious goals, 5G technology imposes stringent requirements that are critical for its deployment and operational success. These include ultra-low latency, near-perfect reliability, and the ability to handle massive bandwidth demands with minimal interruptions. As 5G networks are expected to support the connectivity of millions of devices simultaneously, the concept of a network sharing paradigm has become central to its architecture. A key innovation in this regard is network slicing, a technique that virtualizes the network into multiple dedicated slices. Each slice is dynamically configured and optimized for specific use cases, such as enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC). This approach enables 5G networks to meet the unique demands of diverse applications while maintaining operational efficiency.

Network slicing transforms the traditional network model by enabling Network as a Service (NaaS), where service providers can offer tailored network capabilities without altering the physical infrastructure. This paradigm shift introduces new challenges related to security, orchestration, and management, especially as the complexity of the network increases. However, through advances in Software Defined Networking (SDN) and Network Function Virtualization (NFV), 5G networks are becoming more flexible, programmable, and resilient. These technologies allow for the creation, automation, and management of multiple network slices, each designed for specific services, from consumer mobile use to advanced industrial applications.

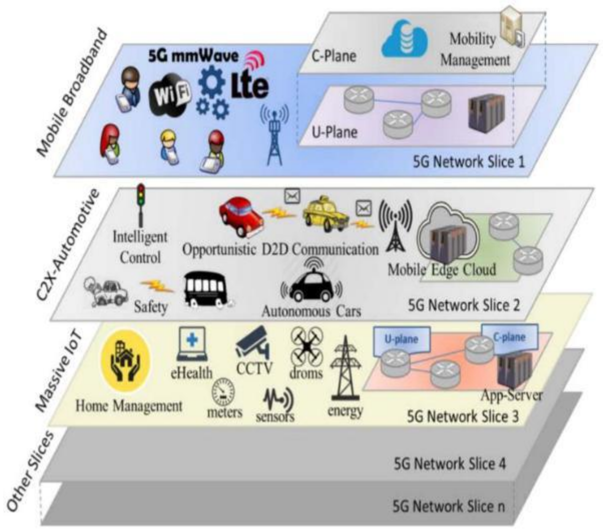

Figure 1: 5G network slices structure [5]

Figure 1 illustrates the structure of 5G network slices, showcasing their ability to operate on a shared physical infrastructure. For example, one slice may be optimized for everyday smartphone communications, ensuring seamless and reliable connectivity. Another slice could be dedicated to supporting Vehicular Ad Hoc Networks (VANETs), providing ultra-fast response times and enhanced reliability, critical for vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications. Additionally, a third slice could focus on the needs of smart cities and Internet of Things (IoT) ecosystems, highlighting the versatility of network slicing in supporting a wide range of technological domains.

As 5G evolves, orchestration and automation are key to managing the lifecycle of these slices—enabling their creation, modification, and removal in real-time, enhancing both flexibility and resource efficiency. This highlights a shift towards network softwarization, where the network becomes more adaptable to specific user requirements and applications. Network Function Virtualization (NFV) and SDN play pivotal roles in achieving this programmable and software-driven network architecture, allowing for rapid deployment of new services and features without the need for extensive physical upgrades.

Recent developments have further solidified the importance of network slicing in future telecom infrastructure. For instance, the integration of artificial intelligence (AI) and machine learning (ML) algorithms into 5G networks has begun to enable predictive network management, ensuring optimal performance and resource allocation across different slices. AI-driven orchestration improves the scalability of the network, allowing for real-time adjustments based on traffic patterns, device densities, and evolving user demands. This is particularly important as 6G research begins to emerge, focusing on even more advanced use cases like holographic communications, tactile internet, and quantum networking, all of which will rely heavily on the dynamic capabilities introduced by network slicing.

The shift to network slicing also paves the way for enhanced security frameworks, where each slice can be isolated and protected based on its specific needs. This multi-level security approach is crucial for ensuring data integrity and privacy, particularly in mission-critical applications such as autonomous vehicles, healthcare, and industrial IoT systems.

In conclusion, the advent of network slicing within 5G signifies a fundamental transformation in the telecommunications landscape. By leveraging the capabilities of SDN and NFV, combined with AI-driven management, 5G networks can deliver unprecedented levels of customization, scalability, and efficiency. This flexibility is essential for supporting the growing demands of smart cities, IoT ecosystems, and beyond, laying the groundwork for a more connected and intelligent future.

III. AN OVERVIEW OF SDN AND NFV

This section delves into two revolutionary network paradigms, Software Defined Networking (SDN) and Network Function Virtualization (NFV), which are pivotal in realizing the concept of network slicing within 5G infrastructures. These technologies facilitate a level of customization and flexibility previously unattainable, aligning perfectly with the diverse and demanding requirements of 5G networks.

A. Software Defined Networking (SDN)

At the heart of traditional networking lie two fundamental technologies: distributed control and transport network protocols, which govern the transfer of data as digital packets across the network's switches and routers. However, SDN emerges as a transformative approach, designed to overcome the inherent limitations of traditional network architectures by decoupling the network's control logic (control plane) from the underlying hardware responsible for data forwarding (data plane). This separation marks a pivotal shift towards a more agile and programmable network infrastructure, where a centralized controller assumes the role of directing traffic across the network, enhancing the efficiency and flexibility of data packet routing.

The key to this paradigm shift is the introduction of programmable interfaces (APIs), with OpenFlow being among the most prominent examples. These APIs facilitate direct communication between the control plane and the data plane, allowing for dynamic network configuration and management. As illustrated in Figure 2, the SDN architecture simplifies the network design by centralizing control functions, thereby enabling more automated provisioning, improved resource management, and faster adaptation to changing network conditions. The adoption of SDN technologies in 5G network slicing is extensively discussed in the literature, reflecting a consensus on its critical role in developing adaptable network slices tailored to specific user needs and service requirements. By leveraging SDN, developers and network operators can design and deploy network slices with the necessary attributes and capabilities to support a wide array of applications and services, from high-bandwidth video streaming to low-latency autonomous vehicle communications.

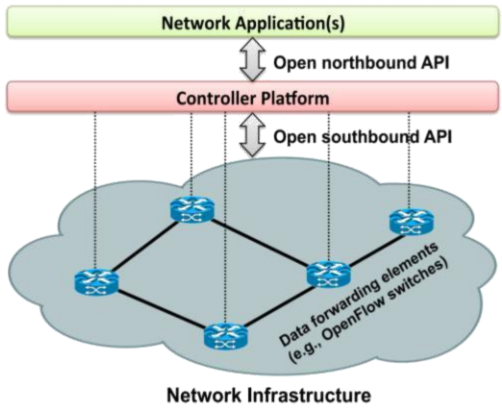

Figure 2 Simplified view of an SDN architecture [8]

B.Network Function Virtualization (NFV)

NFV is another technology leverages virtualization Complementing SDN, NFV represents another cornerstone technology in the evolution towards 5G network slicing. By abstracting network functions from the physical hardware, NFV enables the deployment of network services that are traditionally bound to hardware appliances (such as firewalls, load balancers, and routers) as software instances running on virtualized infrastructure. This abstraction not only reduces the dependency on proprietary hardware but also introduces unprecedented flexibility in how network services are provisioned and scaled.NFV redefines network architecture into a layered model comprising the NFV Infrastructure (NFVI), Virtual Network Functions (VNFs), and Management and Orchestration (MANO) layers. This structured approach allows for the dynamic allocation of network resources and the rapid deployment of network services, catering to the fluctuating demands of 5G networks. Furthermore, NFV's compatibility with SDN amplifies its potential, enabling a synergistic relationship where SDN provides the intelligent network control and programmability, while NFV offers the agility and efficiency of virtualized network functions. In combination, SDN and NFV are instrumental in achieving the vision of 5G network slicing, where each slice can be optimized for specific service types, whether it be Enhanced Mobile Broadband (eMBB), Ultra-Reliable and Low Latency Communications (URLLC), or Massive Machine Type Communications (mMTC). This strategic integration of SDN and NFV within 5G networks paves the way for a new era of telecommunications, characterized by unparalleled service flexibility, scalability, and efficiency, heralding a transformative impact on industries and society at large.

IV. 5G ARCHITECTURAL MODEL

Given the nascent state of 5G technology and the ongoing development of its specifications, creating a comprehensive reference architecture to fully describe it remains a work in progress. Despite these challenges, there is a consensus among industry stakeholders on the critical components that must be integrated into the architecture to ensure the successful deployment and operation of 5G services. Notable organizations, including the Third Generation Partnership Project (3GPP), the International Telecommunication Union (ITU), and the 5G Infrastructure Public Private Partnership (5GPPP), have been instrumental in discussing and establishing the standards and specifications that will define 5G networks[13][14].

The existing mobile network architecture, primarily designed to support traditional services like voice communications and Mobile Broadband (MBB), falls short of meeting the diverse and demanding requirements envisioned for 5G. The ITU categorizes 5G capabilities into three main service types: Enhanced Mobile Broadband (eMBB), Ultra-Reliable and Low Latency Communications (URLLC), and Massive Machine Type Communications (mMTC)[15]. eMBB focuses on catering to the demand for high-bandwidth applications, such as ultra-high-definition video streaming and virtual reality. URLLC, on the other hand, is tailored for latency-sensitive applications, critical for scenarios like Vehicular Ad Hoc Networks (VANETs) and remote surgery, where instantaneous communication is paramount. mMTC is designed to support a high density of connections, essential for applications in smart cities and IoT ecosystems, where vast numbers of devices communicate simultaneously.

The shift towards these new service categories necessitates a departure from traditional network architectures, demanding a more flexible and scalable framework. This new architecture must be capable of dynamically adjusting its resources and functionalities to meet the specific requirements of various applications and services. To address these needs, this section proposes a structured model to delineate the architectural building blocks of 5G networks. This model will illustrate how different network components interact and cooperate, laying the groundwork for a holistic understanding of 5G network architecture. It begins with an abstract representation of the system, identifying key elements and their interconnections, setting the stage for the detailed design of a reference architecture.

Furthermore, the section will explore significant use cases that the 5G system aims to support, along with the system actors involved in these scenarios. This discussion will not only highlight the versatility and potential of 5G technology but also emphasize the critical role of network slicing, SDN, and NFV in facilitating a wide range of applications—from high-speed internet access and immersive entertainment to autonomous vehicles, smart infrastructure, and beyond. By understanding these use cases and the corresponding system actors, we can appreciate the complexities and challenges involved in realizing the full potential of 5G, guiding the development of a network architecture that is robust, adaptable, and capable of transforming the landscape of telecommunications and digital services.

*A. **Struct model***

The proposed reference architecture for a 5G network is conceptualized across four primary layers, accompanied by several control units orchestrating the entire infrastructure. This multi-layered architecture, depicted in Figure 3, serves as the backbone for a highly efficient and flexible 5G network

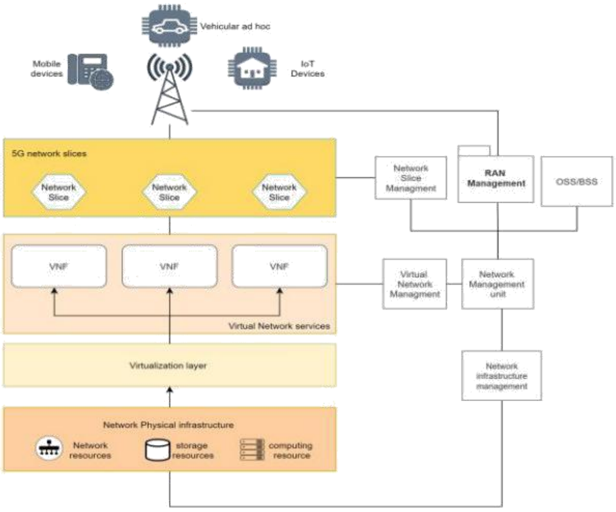

Figure 3 RA model for standalone 5G networks

1. Physical Infrastructure Layer
This foundational layer hosts all the physical devices critical for network management and control, characterized by its distributed nature. It encompasses various networking, storage, and computing resources. The inclusion of Passive Optical Network (PON) technology is noteworthy here, facilitating the management of massive backhaul connections and the linkage between the Next Generation Core (NGC) and Network Radio (NR) active antennas. While several methods exist for 5G transport, such as flexible ethernet, Optical Transport Networks (OTN), and point-to-point fiber optics, PON is distinguished for its point-to-multipoint topology. This topology is not only conducive to fixed access services but also boasts widespread global deployment, making it a superior choice for 5G transportation[16].

2. Virtualization Layer
At this juncture, the physical components of the network undergo virtualization, transforming traditional network functions (like load balancing and switching) into software-based entities. This process is managed by a hypervisor, overseeing the conversion to Virtual Network Functions (VNFs). This decoupling from dedicated hardware enables the implementation of network functions in the cloud, promoting a modular and scalable network infrastructure. The management of these VNFs, alongside the orchestration of various functions into network slices, is handled by a virtual network management unit. This unit's ability to dynamically allocate resources across slices facilitates the creation of customized network slices tailored to specific needs, be it eMBB, mMTC, or URLLC, enhancing the network's adaptability and efficiency[19][20].

3. Network Slice Management Unit (NSM)
The NSM plays a critical role in managing network slices, overseeing aspects such as orchestration, monitoring, security, and assurance. Within the NSM, components are categorized into four main groups:

Orchestration: Manages the lifecycle of network slices from preparation to deployment and oversees the subnetworks within each slice.

Security: Includes user role management, account management, and licenses, ensuring that network operations remain secure and compliant.
Monitoring: Ensures that network slice performance meets predefined thresholds, incorporating fault detection and resource performance tracking.
Assurance: Focuses on data collection, analysis, and recovery management, maintaining the reliability and integrity of the network.
This detailed representation of the NSM, illustrated in Figure 4, highlights the comprehensive approach to managing network slices, each designed to fulfill the diverse requirements of different domains.

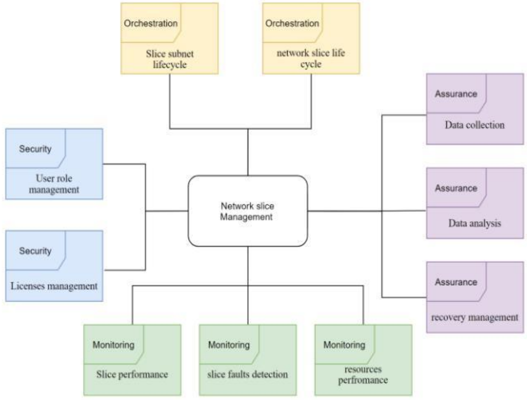

Figure 4 internal components of network slice management unit

Following this exploration of the fundamental components of 5G network architecture, the subsequent subsection will delve into the main use cases facilitated by this system. This understanding not only emphasizes the architectural versatility required to support a wide range of applications but also underscores the importance of a robust and scalable framework capable of adapting to the evolving demands of 5G networking.

B. Dynamic model

In the exploration of 5G network architectures, the Dynamic Model stands as a cornerstone for depicting the network's operational mechanics, focusing on use cases and the interactions between different actors within the system. This model is instrumental in understanding the real-world applications and operational flow of the network, facilitated through the use of Unified Modeling Language (UML) diagrams. These diagrams provide a comprehensive visual representation of the system's functionality, illustrating the full spectrum of interactions between users (actors) and the system under various scenarios, and outlining the system's responses to user requests[22].

In the realm of 5G networks, a diverse array of actors plays pivotal roles, each contributing to the network's overall functionality and service delivery:

5G Network Provider: The primary entity responsible for offering networking services to end-users, ensuring connectivity and network availability.
Network Slicing Broker: Acts as an intermediary, facilitating communication between NFV/SDN providers and network providers to allocate and manage network slices effectively.
NFV/SDN Providers: Specialized entities that construct the virtualized components of the network, leveraging NFV and SDN technologies to enhance the network's flexibility and efficiency.
. The UML use case diagram (Figure 5) delineates the primary use cases inherent to 5G network environments, showcasing the dynamic interactions and processes that define the network's operation:

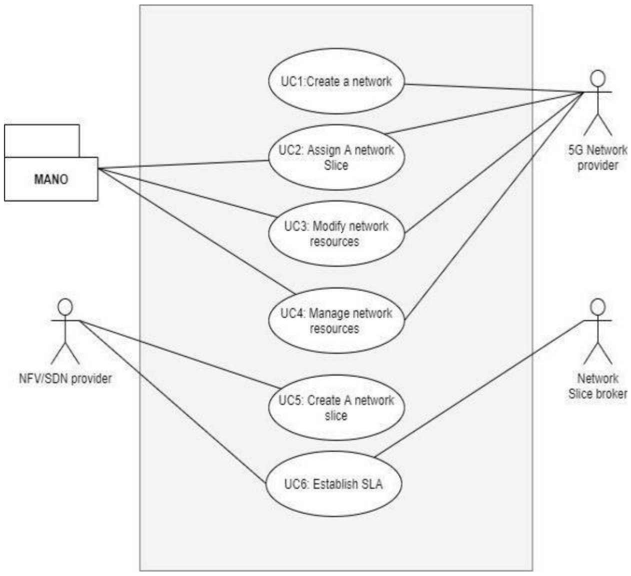

Figure 5 5G network environment UML use cases

UC1: Create a Network
This use case involves creating distinct network slices for different services or applications. Each network slice is tailored to a specific service requirement, such as high bandwidth for mobile broadband (eMBB) or low latency for mission-critical applications (URLLC). For example, a telecom operator could create separate slices for public safety communications and consumer mobile internet, both running on the same physical infrastructure.

Example: A telecom provider can create an ultra-reliable slice for emergency services, ensuring that first responders have uninterrupted access to communication tools even in high-demand scenarios like natural disasters.

UC2: Assign a Network Slice
This use case involves dynamically assigning different network slices to various services or customers. The network provider can allocate resources to specific slices based on real-time demand or priority. For instance, during a live event with high video streaming traffic, the provider may assign more resources to eMBB slices to maintain video quality.

Example: During a global sporting event like the Olympics, broadcasters can be assigned dedicated slices to ensure seamless live streaming of the event, while the public is provided with other network slices for general connectivity.

UC3: Modify Network Resources
Over time, network resource demands change, requiring dynamic modification of allocated resources. This use case involves adjusting the capacity and performance of network slices in real-time to maintain optimal service quality. The ability to modify network resources enables better network efficiency and user satisfaction.

Example: In a smart city, as traffic increases during peak hours, resources could be reallocated from lower-priority applications to transportation systems to ensure that smart traffic lights and public transit communications operate smoothly.

UC4: Manage Network Resources
Effective resource management is central to 5G architecture. This use case focuses on the continuous monitoring and management of resources within and across network slices. Automation plays a key role in ensuring that network components are used efficiently, preventing resource underutilization or overloading.

Example: An industrial IoT network monitoring system in a factory can manage network resources autonomously, ensuring that critical manufacturing processes get higher priority during peak hours, while other non-essential processes use fewer resources.

UC5: Create a Network Slice
Creating a new network slice is central to 5G's flexible and customizable architecture. This use case involves configuring a new slice to meet specific user or service requirements, such as low latency for autonomous vehicles or high throughput for video streaming.

Example: A dedicated network slice could be created for connected vehicles in a smart city. This slice would ensure real-time communication between vehicles and infrastructure, allowing for safe and efficient traffic flow, critical in autonomous driving systems.

UC6: Establish a Service Level Agreement (SLA)
Service Level Agreements (SLAs) ensure that network slices perform according to predefined metrics like uptime, latency, and bandwidth. This use case involves the establishment of SLAs between network providers and service customers to guarantee performance, security, and quality of service.

Example: A hospital using a 5G network for telemedicine would require a strict SLA for its network slice, ensuring ultra-reliable, low-latency communication for remote surgeries or patient monitoring systems.

These use cases, articulated through the UML use case model, provide a structured view into the operational dynamics of 5G networks. They highlight the network's capacity for customization, scalability, and efficient resource management, showcasing the innovative potential of 5G technology to meet the diverse needs of modern digital societies.

C. Key Architecture Features

In addition to these dynamic use cases, the 5G architecture incorporates several advanced features to enhance its overall performance:

- **Software Defined Networking (SDN)**: By decoupling the control and data planes, SDN enables centralized network management, providing the flexibility to reconfigure the network in real-time based on demand and traffic patterns. This is particularly important for managing multiple slices in parallel, each with different performance requirements.

- **Network Function Virtualization (NFV)**: NFV virtualizes traditional network services such as firewalls and routers, making them more flexible and scalable. This reduces dependency on physical hardware and allows for the rapid deployment of network slices based on specific service demands.

- **Artificial Intelligence (AI) and Machine Learning (ML) Integration**: AI and ML play an increasingly critical role in managing 5G networks. These technologies enable predictive analytics, optimizing resource allocation, detecting faults, and improving overall network efficiency by dynamically adjusting network parameters to meet changing conditions.

V. FUTURE DIRECTIONS

As we delve deeper into 5G network research, developing a Reference Architecture (RA) for Non-Standalone (NSA) 5G networks is emerging as a key area of focus. This research aims to apply pattern-oriented analysis to the architecture's components, where patterns are defined as tailored solutions to recurring issues within a specific context[23]. The application of these design patterns to areas such as the Radio Access Network (RAN) management unit is largely unexplored and could reveal significant insights into network management and optimization. Exploring patterns in the slice management unit also offers considerable research potential, enhancing our understanding of 5G architecture and aiding in the creation of a more comprehensive RA. This foundational architecture could be further enhanced by incorporating security and misuse patterns to evolve into a Security Reference Architecture (SRA), which would bolster network defenses against cyber threats. Extending this work to include such patterns not only promises to refine our architectural blueprint for 5G networks but also opens up numerous research opportunities, setting the stage for a more secure, efficient, and resilient 5G infrastructure. Such advancements could drive further innovation in telecommunications, envisioning a future where 5G networks deliver unparalleled service quality, security, and user experience.

In summary, investigating design and security patterns within 5G network architecture is a promising avenue for future research, enhancing the structural foundation of 5G networks and contributing to a secure, scalable, and efficient telecommunications infrastructure. This exploration is poised to unlock new performance and security levels, marking a significant phase in the evolution of 5G technology.

VI. CONCLUSION

In the journey through this paper, we've delved into the nascent yet rapidly evolving realm of 5G technology—a cornerstone for the next wave of breakthroughs in smart cities, the Internet of Things (IoT), and Vehicular Ad Hoc Networks (VANET). The deployment of Non-Standalone (NSA) 5G networks, with their complex interplay of internal and external components, underscores the imperative for a profound understanding of 5G architecture. It is in this context that Reference Architectures (RAs) emerge as invaluable tools, offering a structured lens through which to examine and comprehend the intricate designs of 5G systems. We've embarked on an exploratory voyage, mapping out the essential components that form the backbone of NSA 5G networks—networks characterized by their segmentation into functional slices, each tailored to meet distinct service requirements. This architectural blueprint not only facilitates a granular approach to network service provision but also aligns with the dynamic needs of a diverse user base. Moreover, our discussion has extended to the pivotal use cases and actors within the 5G ecosystem, highlighting the technology's versatility in accommodating varying requirements across a spectrum of applications. The transition towards software-centric network infrastructures, while opening avenues for flexibility and customization, also introduces additional layers of complexity in network management. This shift warrants a cautious approach, particularly concerning the heightened exposure to security vulnerabilities. Nonetheless, the essence of NSA 5G architecture—its ability to offer differentiated services through distinct network slices—stands as a testament to the architectural sophistication and innovation inherent in 5G technology. In concluding, this research not only sheds light on the foundational elements of NSA 5G networks but also underscores the significance of RAs in deconstructing and analyzing such advanced telecommunications systems. The insights gleaned from this study pave the way for future research, especially in enhancing the security framework and operational efficiency of 5G networks. As we stand on the cusp of this technological revolution, the promise of 5G to serve as the backbone for a myriad of future innovations is not just aspirational but well within our grasp. The road ahead is paved with challenges, yet it is replete with opportunities for groundbreaking contributions that will shape the future of telecommunications and digital innovation.

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